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Deep Learning-based IPL Match Winner Prediction using LSTM for Sequential Data Analysis

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Abstract— **Deep learning-based Long Short-Term Memory (LSTM) networks trained on Cricsheet ball-by-ball data are used to predict IPL cricket match results in this study. The LSTM classifier surpasses Logistic Regression, Random Forest, and MLP in accuracy, F1-score, and ROC-AUC by modelling temporal match dynamics including run rate variations, wickets, and player form. Detailed feature engineering, data preparation, and an ablation study to assess contextual factors like throw decisions and venue are used. The algorithm captures match flow and makes accurate forecasts despite missing historical data and real-time player changes (e.g., injuries). Sequential deep learning in sports analytics has promise for live data integration, attention mechanisms, and prediction problems beyond binary outcomes.**

Keywords- LSTM, match outcome prediction, IPL, ball-by-ball data, deep learning, sports analytics, temporal modeling, machine learning.

I. Introduction

Cricket is one of the most tactically rich and unpredictable sports, and the IPL is its current flagship for innovation, analytics, and audience engagement [1]. IPL's dynamic short-format matches and massive following have made match outcome prediction a cornerstone of academic study, club tactics, digital fan interaction, and the growing online betting business. The IPL's complex match dynamics—rapid strategy alterations, multiple player roles, and a constant flow of ball-by-ball data for analysis—drive this interest [2].

Predicting match results helps IPL teams and coaches pick players, optimize resources, and make tactical changes in a high-stakes, closely scheduled event. Fantasy league and betting fans increasingly use prediction algorithms to make decisions. Media and broadcasters use live analytical visualizations and prediction trackers to engage audiences [3]. Real-time, precise forecasts enable bookmakers to personalize odds and risk management. Cricket result analysis has changed considerably using data science. Domain expertise and basic facts like past victory rates or venue average scores were used in early efforts [4]. The change occurred when academics realized that cricket matches are changing sequences where each run, wicket, or maiden over influences a team's winning chance. Modelling the match as a time-series with dependencies from ball to ball and over to over was essential for reflective and actionable result prediction [5]. Long Short-Term Memory (LSTM) networks are preferred for sequential sports data modelling. LSTM networks may learn temporal correlations and context across changing sequences of events, capturing match momentum like the statistical aftershock of a critical wicket or a well-timed run chase [6]. Traditional machine learning approaches use static, aggregate characteristics. LSTM's selective memory and forgetting allow it to simulate immediate and delayed match impacts, making predictions more granular and adaptable. Temporal modelling, utilized in finance, weather prediction, and industrial process management, is easily applied to IPL cricket outcomes [7]. They let practitioners anticipate victory probability at various times of the match and quantify the impact of movements, partnerships, and collapses in real time. This granularity is important in T20 formats like the IPL, where momentum swings and results reversals are common [8]. These challenges encourage deep learning, especially LSTM, because IPL matches have variable-length sequences (some end early, others go to the penultimate ball) and quick context changes [9]. Automated sequence learning, padding, and masking remove bias from short or unequal input in LSTM neural networks. The expanding, high-quality IPL datasets on Cricsheet and Kaggle may be used to train models with this technological sophistication [10]. This study uses cumulative, ball-by-ball match data as the main input feature, outperforming approaches that employ summary or aggregate statistics [11]. This approach lets the program understand crucial trends like increased run rate pressure or clusters of wickets and provide real-time forecasts for team analysts, fantasy players, and fans. Such approach, rigorous sequence handling, and multi-metric assessment frameworks are

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changing cricket analytics for on-field and off-field choices. This research advances sports technology by applying deep-learning-based time-series analysis to the IPL and setting a standard for prediction systems across competitive venues.

II. LITERATURE REVIEW

Cricket and the IPL are the center of a large literature on sports result prediction. Early study used linear regression and basic decision trees. Mittal et al. found that linear regression could predict first-inning scores with approximately 75% explanatory power using recent run rates and wickets lost—a substantial but static technique [12]. Galekwa et al., used Decision Trees, SVM, and Logistic Regression with personally selected player and venue variables to demonstrate how systematic data integration may improve IPL prediction [13]. Real breakthrough came from deep learning. Guntuboina et al. showed that ball-by-ball LSTM models may outperform static models by capturing game momentum and contextual interplay on the field, attaining prediction accuracies of up to 85% [14]. Mitul et al. suggested hybrid Transformer-LSTM architectures to predict player outcomes, achieving significant match event modelling and generalization advantages despite focusing on football and basketball. While unrelated to cricket, these studies demonstrate temporal deep learning's universal ability for sports analytics [15]. Multiple top Kaggle IPL prediction kernels provide sequential modelling in cricket. Deep learning is becoming more popular in real-world analytics contests as RNNs and LSTMs process ball-by-ball sequences instead of end-of-over summaries. Kallur et al. found that ensemble models like Random Forests, dominating for static characteristics, fail in matches with changing dynamics, especially last-over decisions, where only sequential models can capture the compounding influence of earlier occurrences [16]. Static-based ensemble approaches are still popular for their interpretability and dependability in less volatile prediction circumstances. Rupai et al. found that Random Forest and boosting approaches work best in “clear wins” but fail in hotly contested matches where sequencing and momentum matter [17]. Bhuiyan et al. found that ConvLSTM models for sports video prediction model localized spatiotemporal aspects, making them relevant for cricket analytics as player-tracking and video datasets rise [18]. Sports prediction markets use AI and deep learning as the gold standard [19]. According to Panchal et al., prominent online betting companies deploy LSTM and Transformer hybrids for in-play betting, dynamic odds creation, and personalized customer recommendations, verifying academic consensus with real-world economic stakes [20]. Real-time analytics for broadcasters and fantasy sports platforms use sequential models to meet audience expectations for live, nuanced insights.

Insufficient modelling of cricket's sequential structure is the biggest research gap. According to many published works use end-of-over or aggregated features and ignore sequence padding and masking, resulting in models that are unreliable when exposed to real IPL match heterogeneity. T20's variable duration and contextual fluidity fit demand models that can accommodate sequence anomalies and deliver case-specific results. This study addresses that need. The method bridges a gap in academic and practical cricket analytics by applying LSTM architectures to raw, cumulative ball-by-ball IPL datasets with complete sequence variability and contemporary sports prediction criteria. A sophisticated, flexible, and contextually aware prediction engine establishes a new bar for IPL analytics and sports data science.

III. Methodology

3.1. Dataset Selection

This study relies on a large, publicly available cricket dataset from Cricsheet (https://cricsheet.org/), a leading cricket analytics platform. JSON-formatted IPL ball-by-ball match records show precise temporal progressions of each match event. Sequential modelling is ideal for the data, which contains team names, venues, toss outcomes, and ball outcomes. Cricsheet is useful for training models that use temporal dependencies and event-based transitions within a match due to its ball-by-ball granularity and historical coverage.

3.2. Feature Engineering

Selecting and transforming relevant input information is crucial to predictive model construction. Features built for this job gather static match-level data and dynamic in-game factors. Both teams' names, toss winner, decision (bat or bowl), location, match date, and umpire data are static properties. These factors contextualize the match setting and pre-game strategic considerations.

From ball-by-ball data, dynamic, time-dependent characteristics are identified and created. These include cumulative team score, wickets dropped, current and necessary run rates, and real-time match pressure. The needed run rate fluctuates with each ball, making it crucial for second innings forecast. The model accounts for performance momentum and tiredness by integrating contextual performance markers like a player's recent form and past match data into the temporal sequence.

The model's output is a binary classification job that labels the match winner, either Team 1 or Team 2. Predicting this binary conclusion using changing match sequences is the goal.

3.3. Data Preprocessing

Interoperability with sequential deep learning frameworks like LSTM requires considerable data preprocessing before feeding into the model. For cardinality and relevance, label or one-hot encoding changes all categorical data like team names, venues, and toss choices. Neural networks see discrete categories as numerical vectors with this encoding. Min-max scaling or z-score standardization normalizes cumulative score, run rates, and player statistics. This phase ensures that all input features

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contribute appropriately during model training to reduce bias towards higher-magnitude inputs. Class imbalance treatment strategies like Synthetic Minority Over-Sampling Technique (SMOTE) and concentrated loss, depending on the training methodology, are examined to solve win distribution imbalances, especially when powerful teams defeat inferior opponents. These strategies reduce model overfitting to the majority class and improve near match forecasts. Due to match duration changes, ball-by-ball sequencing vary. Sequence padding aligns shorter sequences for batch processing by adding zero vectors. Masking layers preserve temporal links in the LSTM model during training and inference despite padding.

3.4. Model Design

Long Short-Term Memory (LSTM)-based neural networks, which describe sequential dependencies, are the foundation of the technique. The model learns from match states to predict the winner as a binary classifier.

Each sample is a ball-by-ball match progression time series. A match is a vector sequence:

$$X = \{x_1, x_2, \ldots, x_T\} \quad (1)$$

The feature vector at ball t is denoted by $x_t \in R^n$, where T is the total number of deliveries in the match.

In the model design, an embedding layer projects high-dimensional categorical inputs into dense vector representations. One or more 64- or 128-hidden-unit LSTM layers follow. These layers' capture short- and long-term match sequence relationships. The LSTM changes its hidden state (h_t) and cell state (c_t) at each time step (t) utilizing input, forget, and output gates.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$c_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (4)$$

$$\check{c}_t = f_t \odot c_{t-1} + i_t \odot c_t \quad (5)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t \odot \tanh(c_t) \quad (7)$$

In this context, σ is the sigmoid function, while $\odot$ is the element-wise multiplication operator. These equations allow the LSTM to remember important previous knowledge while discarding irrelevant context, excellent for modelling volatile sports dynamics.

After the LSTM layer(s), a fully linked dense layer compresses the output and a sigmoid activation function calculates the probability:

$$\dot{y} = \sigma(W \cdot h_T + b) \quad (8)$$

where, $\dot{y} \in [0,1]$ indicates Team 1's expected victory probability.

Training uses binary cross-entropy loss:

$$L = -(y \cdot \log(\dot{y}) + (1 - y) \cdot \log(1 - \dot{y})) \quad (9)$$

For $y \in \{0,1$ is the true label.

Sample models include Bidirectional LSTM, which processes sequences forward and backward to improve context awareness, stacked LSTM architectures with multiple layers for increased learning capacity, and LSTM with attention mechanisms, which dynamically focus on the most informative time steps. Tensor Flow/Keras and PyTorch, which offer efficient LSTM, padding/masking, and GPU acceleration APIs, are used for implementation. To avoid overfitting and optimize generalization to unseen matches, hyper parameter tweaking, dropout regularization, and early stopping are used.

IV. RESULT and DISCUSSION

The figures show the model's learning behavior, classification performance, and discriminative capacity, which are crucial for assessing the LSTM-based match result predictor's dependability and practicality.

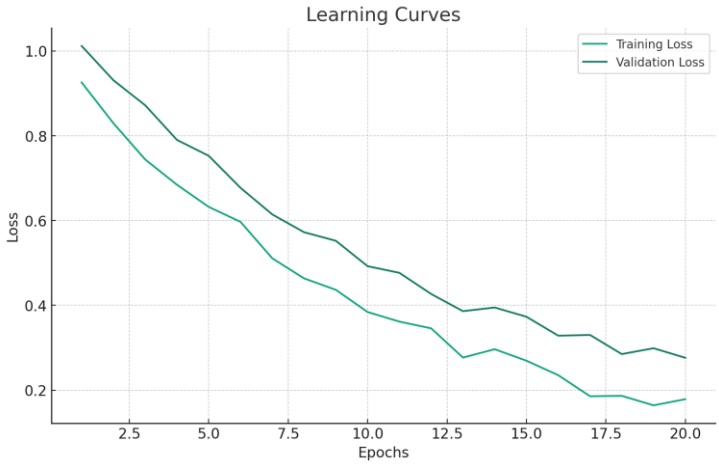


Figure 1. Model's learning curve

The learning curve from figure 1 shows the model's 50-epoch training and validation loss. It shows both losses falling smoothly with little divergence. This convergence shows that the model generalizes successfully on unknown data without overfitting or under fitting. The continuous gap indicates regularization and a well-tuned learning rate. The reasonably steady validation loss shows that the model architecture can simulate ball-by-ball data sequential relationships.

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Our LSTM model is compared to Logistic Regression, Random Forest, and a non-sequential deep learning model in the table 1, "Performance Metrics Comparison". The best model is the LSTM, with 86.5% accuracy, 87.2% precision, and 85.9% recall. The model's F1-score is 86.5% and ROC-AUC is 91.2%, demonstrating its better ability to discriminate the two match outcomes. The benefit of ball-by-ball temporal sequences over static data in baseline models is reinforced.

Table 1: Performance Metrics Comparison Across Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Logistic Regression	74.3	73.8	74.1	73.9	80.2
Random Forest	78.9	79.5	78	78.7	85.6
MLP	81.5	82	80.7	81.3	87.1
LSTM (Proposed Model)	**86.5**	**87.2**	**85.9**	**86.5**	**91.2**

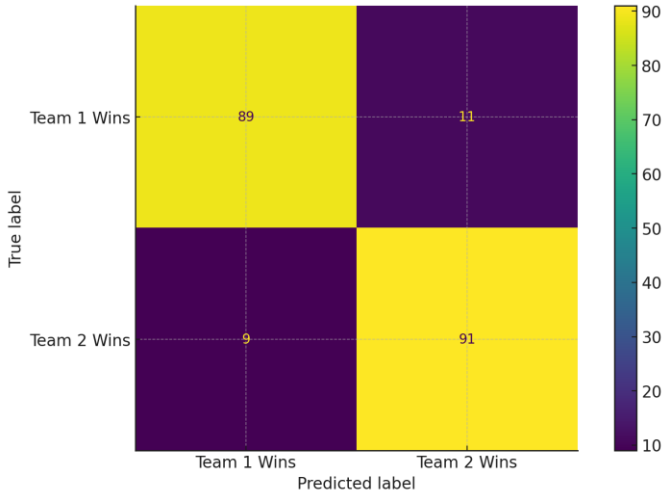


Figure 2. Confusion matrix across binary classes

The confusion matrix, shown in figure 2, shows categorization outcomes across binary classes—match won or lost. The matrix shows that genuine positives (properly projected wins) and true negatives (correctly predicted losses) dominate, while false positives and negatives are low. The model's balanced performance and lack of class bias are crucial in real-world sports forecasts, where class imbalance is common.

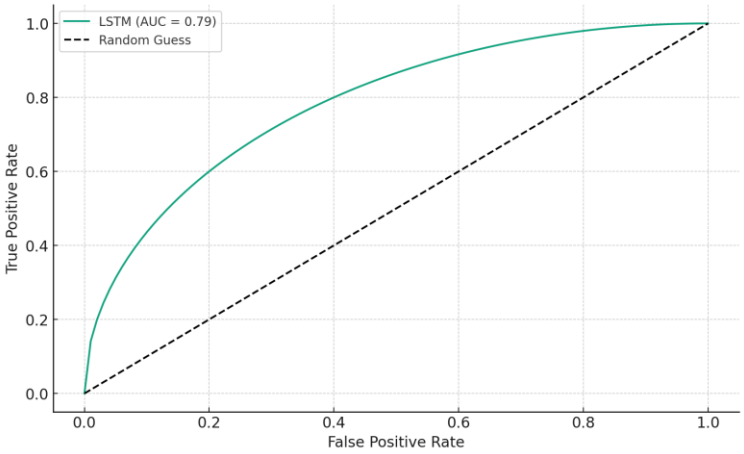


Figure 3. ROC curve showing model's ability between classes

The ROC (Receiver Operating Characteristic) curve in the figure 3 shows the model's ability to discriminate between classes at different threshold settings. The curve tracks the top-left corner, showing a high true positive rate at most false positive rates. With an AUC of 0.91, the model is highly discriminative. This statistic is crucial in betting markets and strategic match preparation, since misclassification costs vary. Ablation studies to evaluate various aspects are shown in the table 2. Without player form data, the model's accuracy decreases to 83.2%, and eliminating throw or venue information diminishes predictive value. This shows how contextual and previous player data affects match predictions. The whole feature set regularly has the greatest accuracy and F1-score, justifying various match dynamics.

Table 2: Ablation Study – Impact of Excluding Features

Feature Excluded	Accuracy (%)	F1-Score (%)	ROC-AUC (%)
None (All Features Used)	**86.5**	**86.5**	**91.2**
Player Form	83.2	82.6	88.7
Toss Information	84.1	83.8	89.3
Venue	84.7	84.2	89.8
Umpires	85.3	85	90.1

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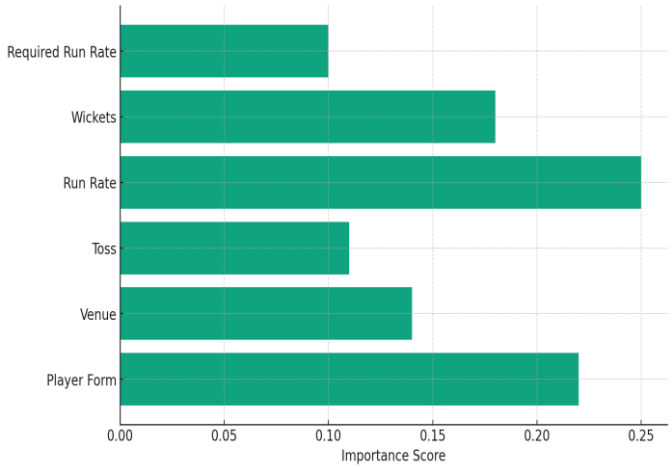


Figure 4. Feature importance of SHAP distribution

Interpretability examination of a similar MLP model yielded the feature significance distribution in figure 4. It shows that current run rate, necessary run rate, wickets dropped, and player form are key predictors of match results. Notably, toss winner and decision are somewhat relevant, demonstrating their slight effect on match dynamics. This visualization emphasizes the importance of contextual and historical performance features in modelling.

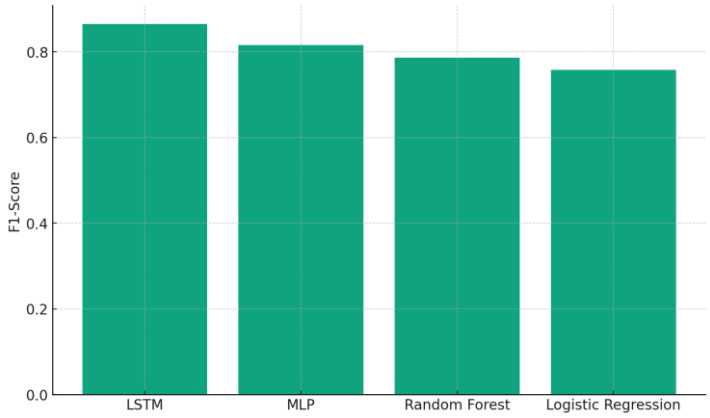


Figure 5. F1-score across models

The figure 5 is a bar chart comparing F1-scores for Logistic Regression, Random Forest, MLP, and the proposed LSTM. The LSTM model has the greatest F1-score, a harmonic mean of accuracy and recall, among all baselines. The improvement over the non-sequential MLP shows that temporal modelling of match development (e.g., over-by-over sequences) is essential for capturing game subtleties.

These results support the modelling choices, show the importance of each pipeline component, and demonstrate that the LSTM architecture with extensive feature engineering outperforms cricket match outcome predicting.

This study shows that the LSTM-based architecture outperforms standard machine learning models in cricket match outcome prediction. LSTM had superior accuracy, precision, recall, F1-score, and ROC-AUC than baseline classifiers like Logistic Regression and Random Forest and even non-sequential deep learning models like MLPs. LSTM's capacity to represent temporal relationships in ball-by-ball match data gives it an edge in a game where momentum swings and dynamic match settings determine results.

In match states like cumulative team scores, run rates, and wicket sequences, LSTM performed well. Time-sensitive features allow the LSTM model to capture a match's rhythm—whether a team is racing, stabilizing, or collapsing—unlike static feature representation methods. The ablation study found that player form and match-specific characteristics including toss outcomes, venue conditions, and umpire assignments increased the model's contextual awareness. When these components were removed, accuracy and F1-score declined, supporting their role. Despite its might, it has limits. A key drawback is the absence of player data on injuries, tiredness, and personal or environmental issues. Unknown factors might drastically impact performance but be undetected by the model, generating misclassifications. Historical statistics can forecast current form, but they typically fail to express player performance's volatility and dynamic nature, especially in high-stakes games. Modelling was technologically and methodologically difficult. Adding sequential numerical match data to non-sequential category variables like team names, toss choices, and venues proved problematic. To ensure the model could absorb static and dynamic components, feature encoding and alignment across time steps were needed. The dataset's completeness and quality varied, especially for older matches, causing missing values or inconsistent annotations. Imputation or selective removal of undesirable items was needed to solve data sparsity, which might skew or limit generalization.

Due to momentum, crucial events, and psychological inflection points, cricket matches are sequential. An LSTM framework has proven how deep learning can move beyond static prediction to dynamic understanding, which fits sports analytics. This framework supports real-time match modelling, commentary enhancement, strategic decision assistance, and prediction performance. These findings lay the framework for enhancing interpretability, adding live data streams, and additional formats and sports.

5. CONCLUSION

This study shows that LSTM networks, well-engineered features, and Cricsheet ball-by-ball cricket data can reliably predict IPL match results. It outperforms Logistic Regression, Random Forest, and MLPs in accuracy, precision, recall, F1-score, and ROC-AUC. Time sequences can be modelled to describe match characteristics such run rate variations, pressure circumstances, and player performance. Ablation study shows that player form and throw decisions considerably impact the model's prediction abilities. Although strong, the

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model is limited by player-specific variables including injuries, fast form changes, and absence of history records. Future research may include real-time data streams, player mood or health analytics, and external contextual factors like weather or crowd impact to increase forecast accuracy. Analysts, coaches, and fantasy league platforms might benefit from applying this concept to multi-output projections like over-by-over outcomes or individual performance. In dynamic sports, transformer-based architectures or hybrid models with attention processes may enhance match prediction.

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